

3. The external characteristics of two d.c. shunt generators *A* and *B* are straight lines over the working range between no-load and full-load.

	Generator <i>A</i>		Generator <i>B</i>	
	No-load	Full-load	No-load	Full-load
Terminal P.D. (V)	400	360	420	370
Load current (A)	0	80	0	70

Determine the common terminal voltage and output current of each generator when sharing a total load of 100 A.

(57.7 A ; 42.3 A ; 378.8 V)

4. Two shunt generators operating in parallel have each an armature resistance of 0.02Ω . The combined external load current is 2500 A. If the generated e.m.f.s. of the machines are 560 V and 550 V respectively, calculate the bus-bar voltage and output in kW of each machine. (530 V ; 795 kW ; 530 kW)

5. Two shunt generators *A* and *B* operate in parallel and their load characteristics may be taken as straight lines. The voltage of *A* falls from 240 V at no-load to 220 V at 200 A, while that of *B* falls from 245 V at no-load to 220 V at 150 A. Determine the current which each machine supplies to a common load of 300 A and the bus-bar voltage at this load.

(169 A ; 131 A ; 223.1 V)

6. Two shunt-wound d.c. generators are connected in parallel to supply a load of 5,000 A. Each machine has an armature resistance of 0.03Ω and a field resistance of 60Ω , but the e.m.f. of one machine is 600 V and that of the other is 640 V. What power does each machine supply ?

(1,004 kW ; 1,730 kW including the fields)

7. Two shunt generators running in parallel share a load of 100 kW equally at a terminal voltage of 230 V. On no-load, their voltages rise to 240 V and 245 V respectively. Assuming that their volt-ampere characteristics are rectilinear, find how would they share the load when the total current is reduced to half its original value ? Also, find the new terminal voltage.

(20 kW ; 30 kW, 236 V)

8. Two generators, each having no-load voltage of 500 V, are connected in parallel to a constant resistance load consuming 400 kW. The terminal p.d. of one machine falls linearly to 470 V as the load is increased to 850 A while that of the other falls linearly to 460 V when the load is 600 A. Find the load current and voltage of each generator.

If the induced e.m.f. of one machine is increased to share load equally, find the new current and voltage.

($I_1 = 626 \text{ A}$; $I_2 = 313 \text{ A}$; $V = 479 \text{ V}$; $I = 469.5 \text{ A}$; $V = 484.4 \text{ V}$)

9. Estimate the number of turns needed on each interpole of a 6-pole generator delivering 200 kW at 200 V ; given : number of lap-connected armature conductors = 540 ; interpole air gap = 1.0 cm ; flux-density in interpole air-gap = 0.3 Wb/m^2 . Ignore the effect of iron parts of the circuit and of leakage.

[10] (Electrical Machines, B.H.U. 1980)

OBJECTIVE TEST – 27

- In d.c. generators, armature reaction is produced actually by
 - its field current
 - armature conductors
 - field pole winding
 - load current in armature
- In a d.c. generator, the effect of armature reaction on the main pole flux is to
 - reduce it
 - distort it
 - reverse it
 - both (a) and (b)
- In a clockwise-rotating loaded d.c. generator, brushes have to be shifted
 - clockwise
 - counterclockwise
 - either (a) or (b)
 - neither (a) nor (b)
- The primary reason for providing compensating windings in a d.c. generator is to
 - compensate for decrease in main flux
 - neutralize armature mmf
 - neutralize cross-magnetising flux
 - maintain uniform flux distribution.
- The main function of interpoles is to minimize between the brushes and the commutator when the d.c. machine is loaded.
 - friction
 - sparking
 - current
 - wear and tear

6. In a 6-pole d.c. machine, 90 mechanical degrees correspond to electrical degrees.
 - (a) 30
 - (b) 180
 - (c) 45
 - (d) 270
7. The most likely cause(s) of sparking at the brushes in a d.c. machine is /are
 - (a) open coil in the armature
 - (b) defective interpoles
 - (c) incorrect brush spring pressure
 - (d) all of the above
8. In a 10-pole, lap-wound d.c. generator, the number of active armature conductors per pole is 50. The number of compensating conductors per pole required is
 - (a) 5
 - (b) 50
 - (c) 500
 - (d) 10
9. The commutation process in a d.c. generator basically involves
 - (a) passage of current from moving armature to a stationary load
 - (b) reversal of current in an armature coil as it crosses MNA
 - (c) conversion of a.c. to d.c.
 - (d) suppression of reactance voltage
10. Point out the WRONG statement. In d.c. generators, commutation can be improved by
 - (a) using interpoles
 - (b) using carbon brushes in place of Cu brushes
 - (c) shifting brush axis in the direction of armature rotation
 - (d) none of the above
11. Each of the following statements regarding interpoles is true **except**
 - (a) they are small yoke-fixed poles spaced in between the main poles
 - (b) they are connected in parallel with the armature so that they carry part of the armature current
 - (c) their polarity, in the case of generators is the same as that of the main pole ahead
 - (d) they automatically neutralize not only reactance voltage but cross-magnetisation as well
12. Shunt generators are most suited for stable parallel operation because of their voltage characteristics.
 - (a) identical
 - (b) dropping
 - (c) linear
 - (d) rising
13. Two parallel shunt generators will divide the total load equally in proportion to their kilowatt output ratings only when they have the same
 - (a) rated voltage
 - (b) voltage regulation
 - (c) internal $I_a R_a$ drops
 - (d) both (a) and (b)
14. The main function of an equalizer bar is to make the parallel operation of two over-compounded d.c. generators
 - (a) stable
 - (b) possible
 - (c) regular
 - (d) smooth
15. The *essential* condition for stable parallel operation of two d.c. generators having similar characteristics is that they should have
 - (a) same kilowatt output ratings
 - (b) dropping voltage characteristics
 - (c) same percentage regulation
 - (d) same no-load and full-load speed
16. The main factor which leads to unstable parallel operation of flat-and over-compound d.c. generators is
 - (a) unequal number of turns in their series field windings
 - (b) unequal series field resistances
 - (c) their rising voltage characteristics
 - (d) unequal speed regulation of their prime movers
17. The simplest way to shift load from one d.c. shunt generator running in parallel with another is to
 - (a) adjust their field rheostats
 - (b) insert resistance in their armature circuits
 - (c) adjust speeds of their prime movers
 - (d) use equalizer connections
18. Which one of the following types of generators does NOT need equalizers for satisfactory parallel operation?
 - (a) series
 - (b) over-compound
 - (c) flat-compound
 - (d) under-compound.

ANSWERS

1. (d) 2. (d) 3. (a) 4. (c) 5. (b) 6. (d) 7. (d) 8. (a) 9. (b) 10. (d) 11. (b)
 12. (b) 13. (d) 14. (a) 15. (b) 16. (c) 17. (a) 18. (d)